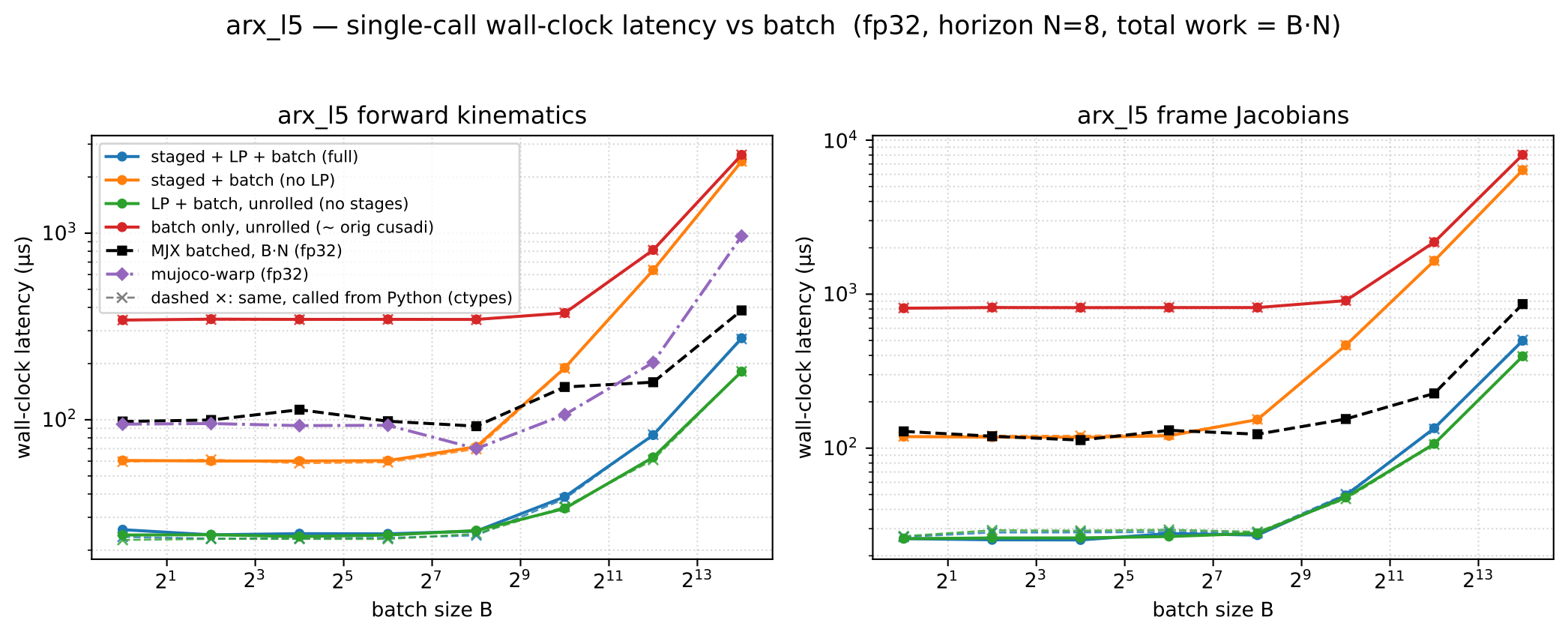
arx_l5 — single-call wall-clock latency vs batch (fp32, horizon N=8, total work = B·N)
arx_l5 forward kinematics
arx_l5 frame Jacobians
staged + LP + batch (full)
staged + batch (no LP)
LP + batch, unrolled (no stages)
batch only, unrolled (~ orig cusadi)
MJX batched, B·N (fp32)
mujoco-warp (fp32)
dashed ×: same, called from Python (ctypes)
wall-clock latency (µs)
batch size B
10^2
10^3
10^4
2^1
2^3
2^5
2^7
2^9
2^{11}
2^{13}